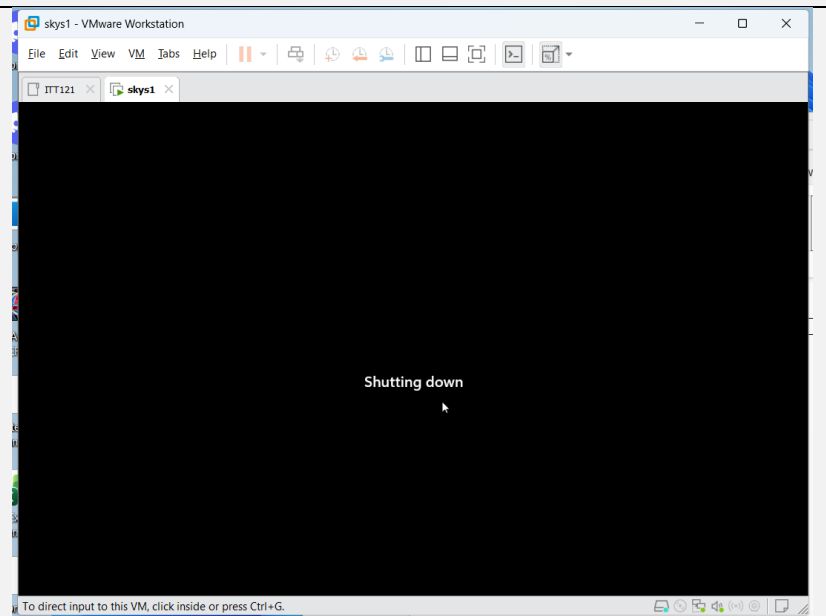
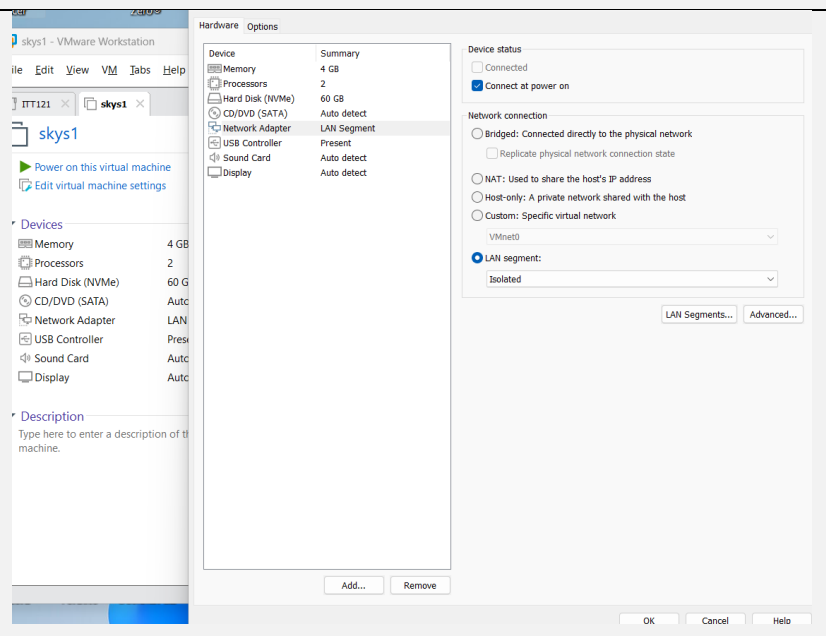


# DHCP

Before we add DHCP to our server we need to isolate our network so our DHCP server does not get out of our machines. To do this we need to shut down the server and the workstation.

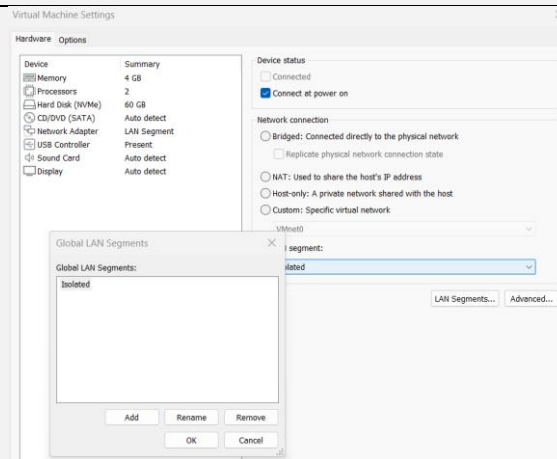


When the machines are off click on edit virtual machine setting and find the network adapter. Check the box “LAN segments” and then click on the box “LAN Segments...”

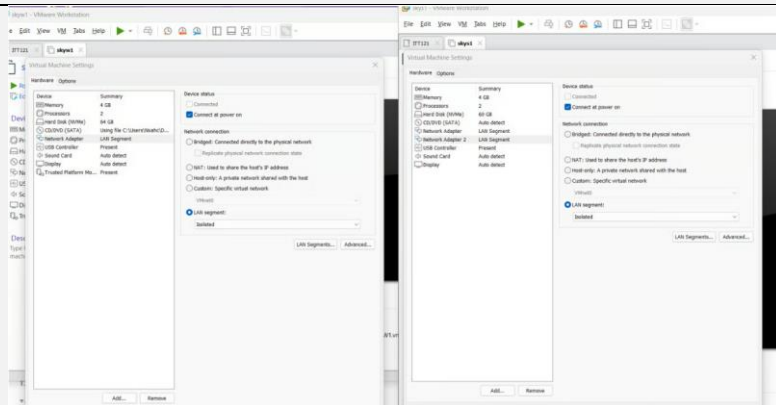


# DHCP

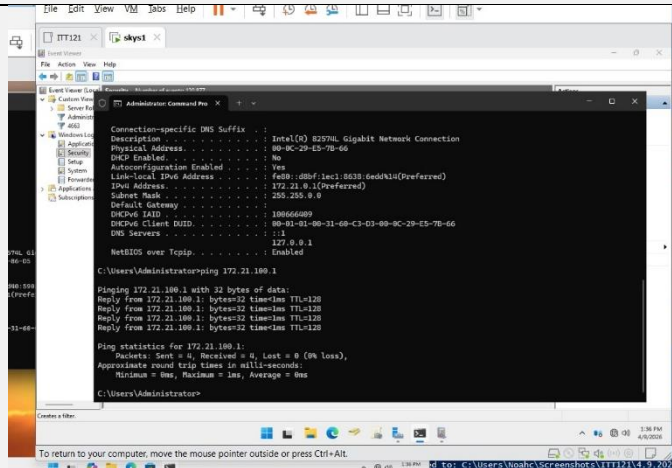
Add a new segments and give it name like Isolated.



For both of the machines make sure that the network adapter is using a LAN Segment and make sure that Isolated is selected for both machines.

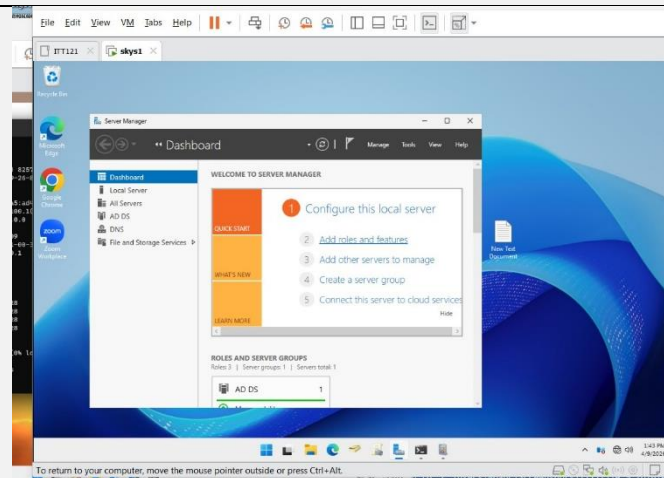


Now start the workstation and server again and test the connection between the two machines by pinging from command prompt. To open command prompt click the windows icon and type in “cmd” and then hit enter on the keyboard.

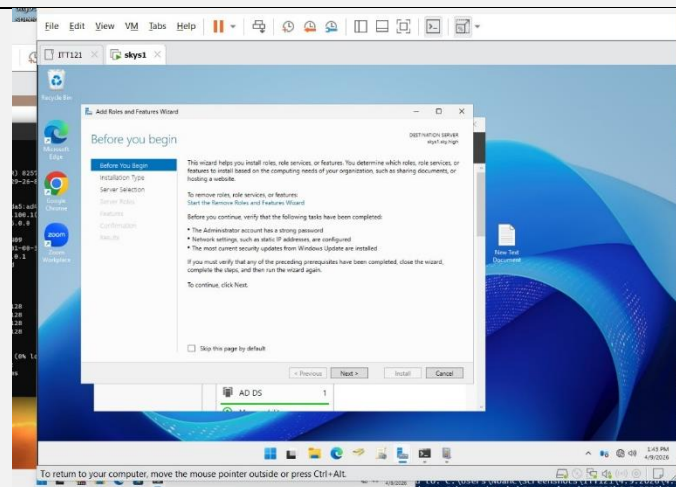


# DHCP

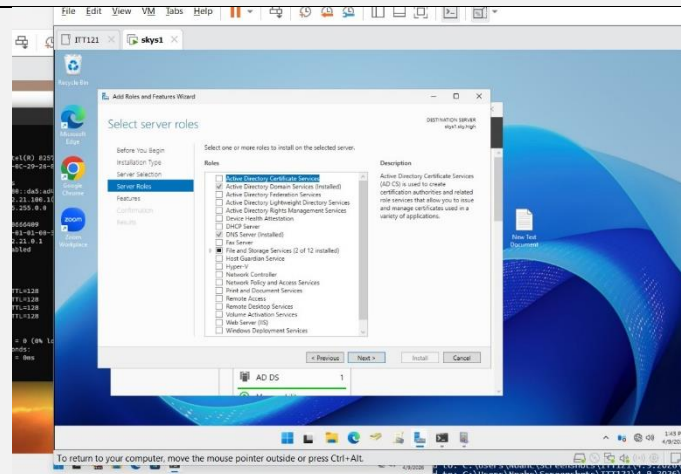
Now in the server bring up the Server Manager and click of “Add roles and features”.



Continue through the pages by hitting next

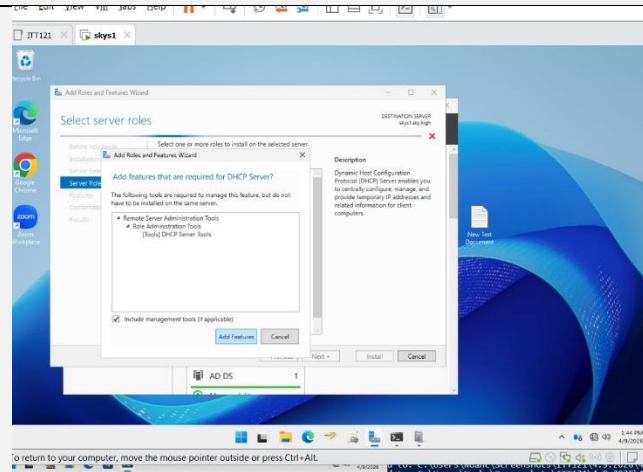


Stop at Select Server Roles and then find and check the DHCP option.

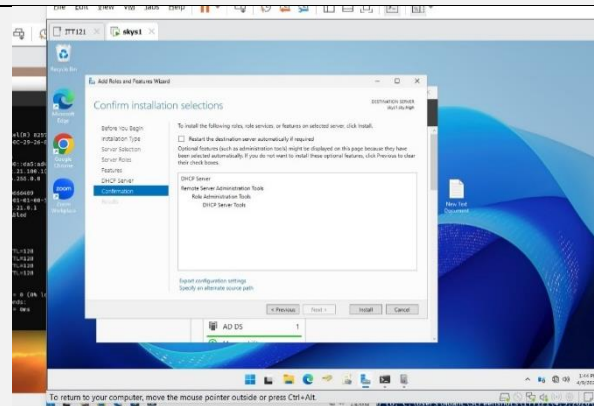


# DHCP

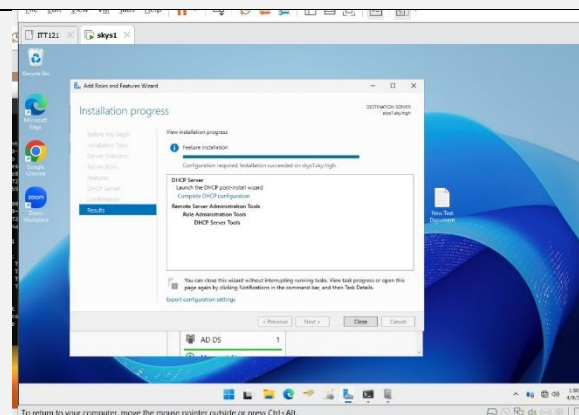
When you hit next you'll see a more detailed list of what you are installing, you can click **Add Features** and continue.



Keep clicking next on the pages until you get the **Confirm Installation Selections** page. Click **install**.

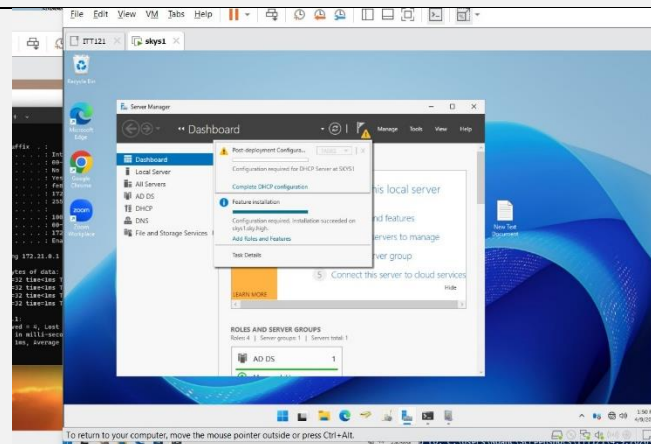


Once the install is done click **close**.

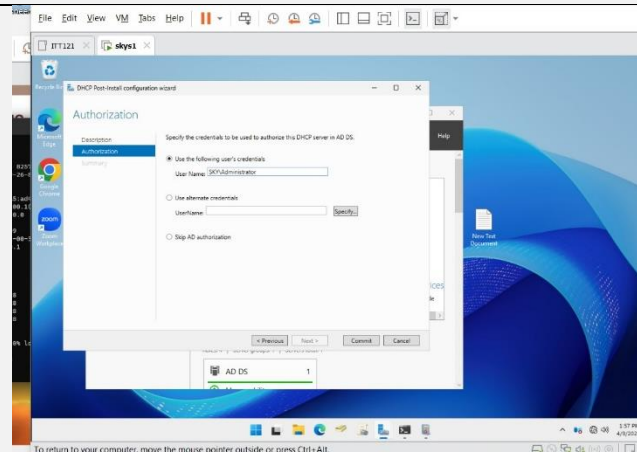


# DHCP

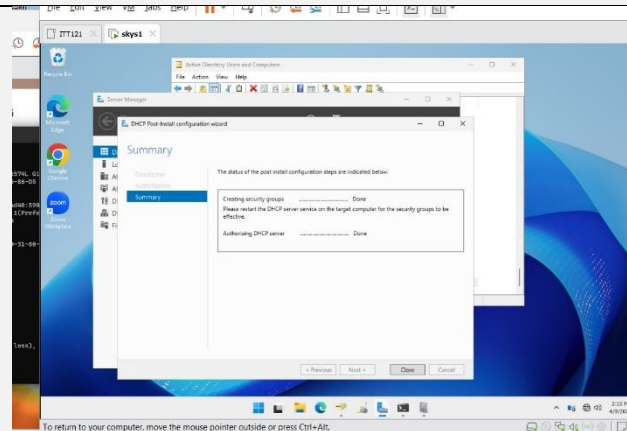
Now in the Server Manager there is a flag in the top toolbar with a caution icon. Click on the flag and then click “Complete DHCP Configuration”



After the description we will have the option to specify the credential to authorize. It would be best practice to not have this as the main Administrator, but with how our current server and workstation are setup we can use the Administrator this time. Click the commit button.

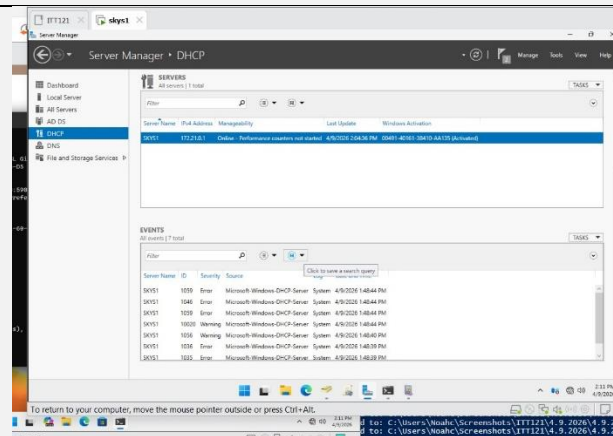


In the summary both tasks should be marked with Done. Click close.

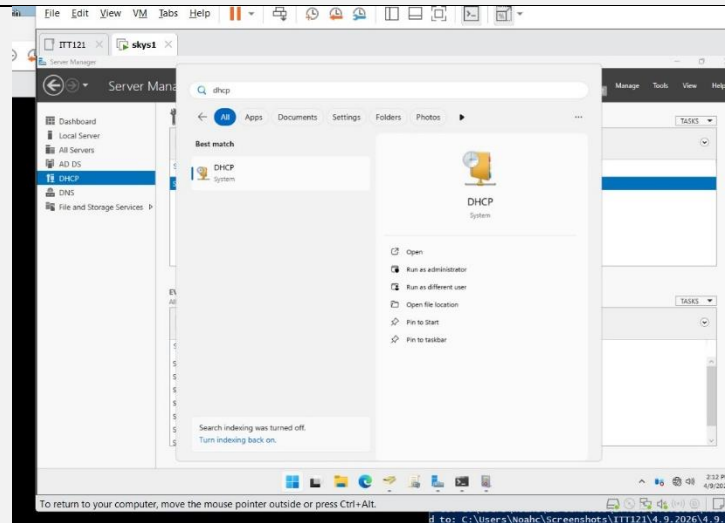


# DHCP

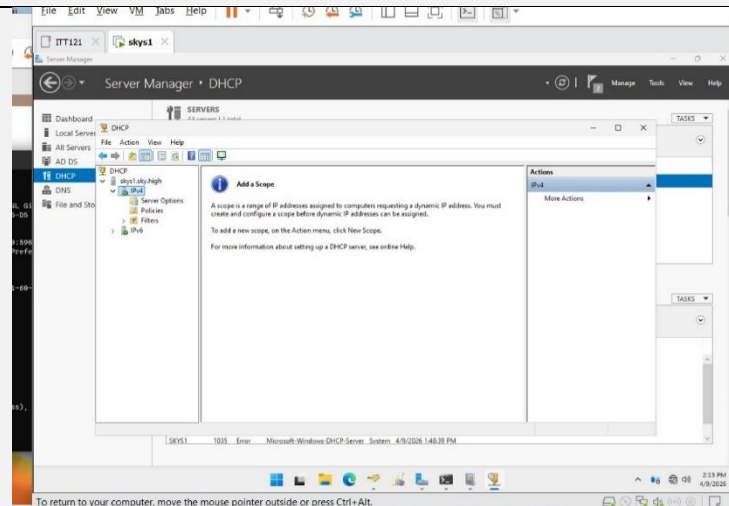
Now in the server manager go to DHCP by clicking on it on the left hand side. You should see that the server is activated under the Windows Activation side.



Now we need to add a scope. Click the windows icon and type in DHCP and then hit enter.

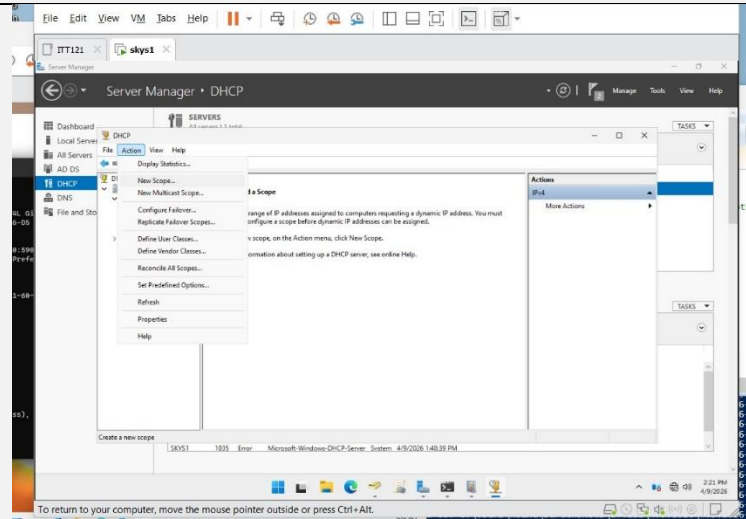


In the DHCP application expand the domain name and find the IPv4 option.

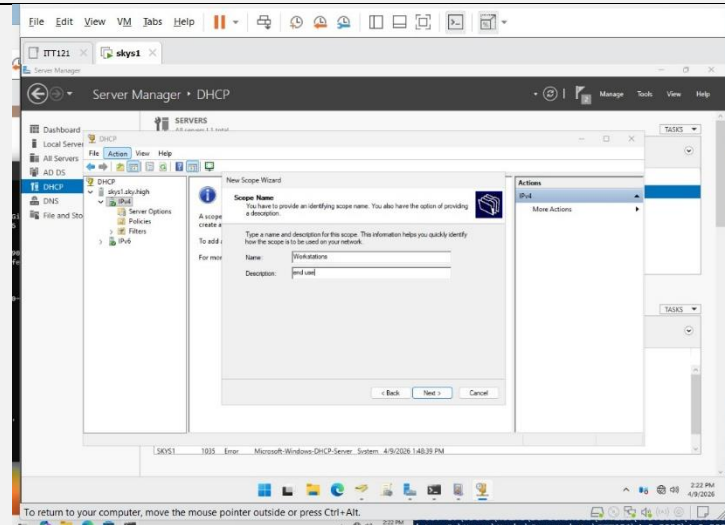


# DHCP

Right click IPv4 and click  
New Scope.



Give the scope a descriptive  
name like “workstations” and  
a description of “end users”  
so we know what this scope is  
for.



# DHCP

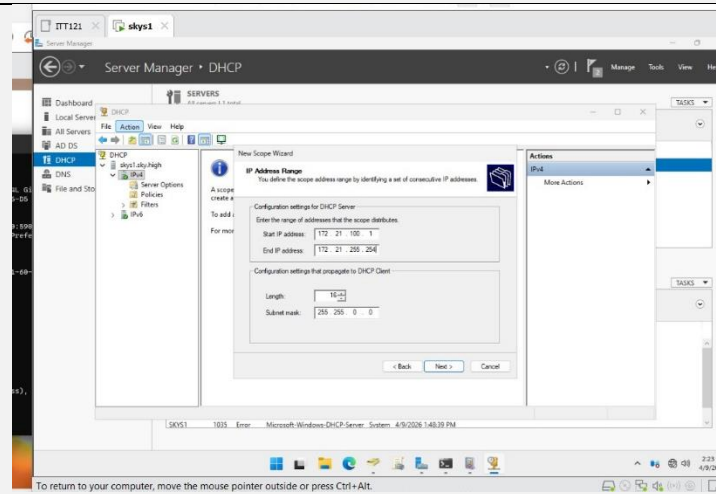
For the IP Address Range the start IP address is 172.#.100.1

The end IP address is

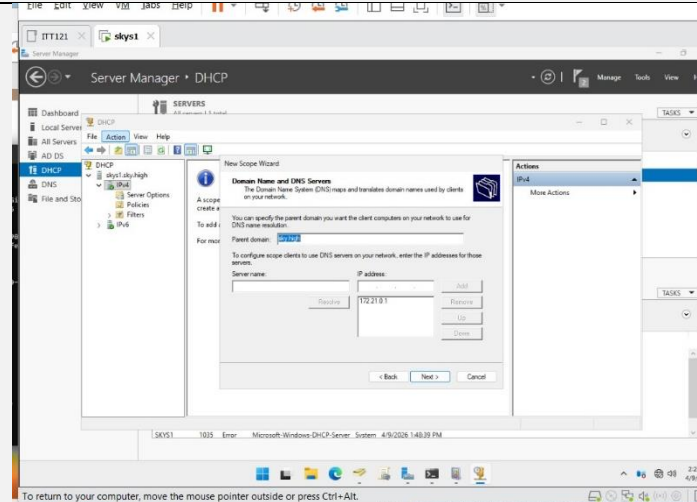
172.#.255.254

Where # is the number you were assigned at the start of the semester. Length is 16 and the subnet mask is 255.255.0.0

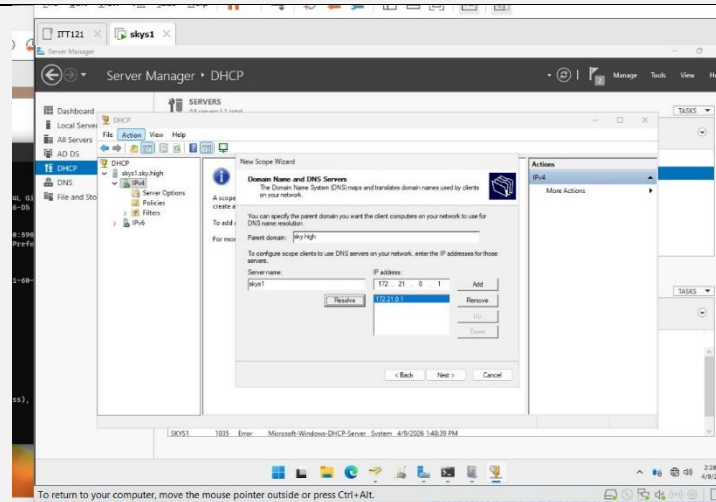
Hit next



Continue through the pages until you are at the Domain Name and DNS Servers page. Here we need to add our dns server.

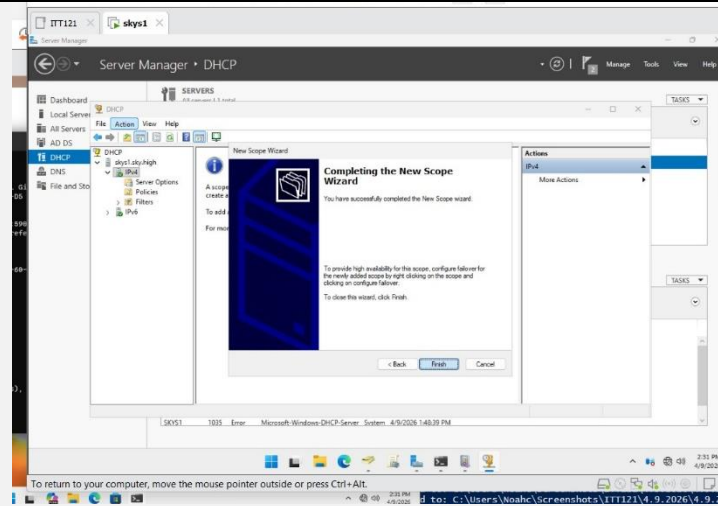


Type in your server name in the server name box and hit Resolve to automatically fill out the IP address or enter the IP address manually. Click add and then next.

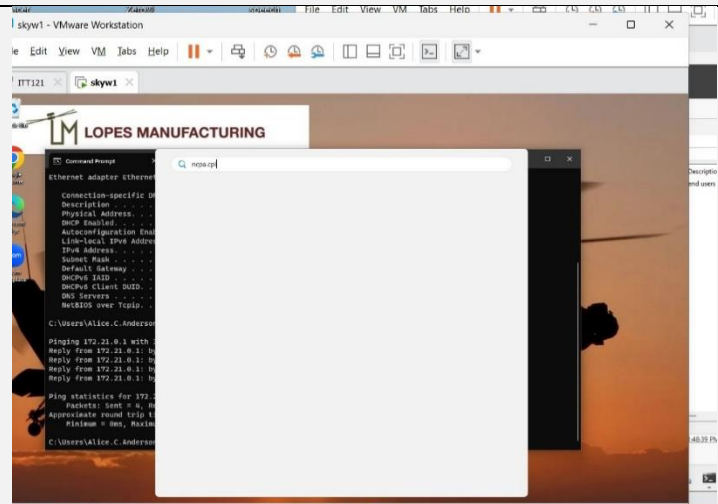


# DHCP

Now we can finish the New Scope setup. Click Finish.

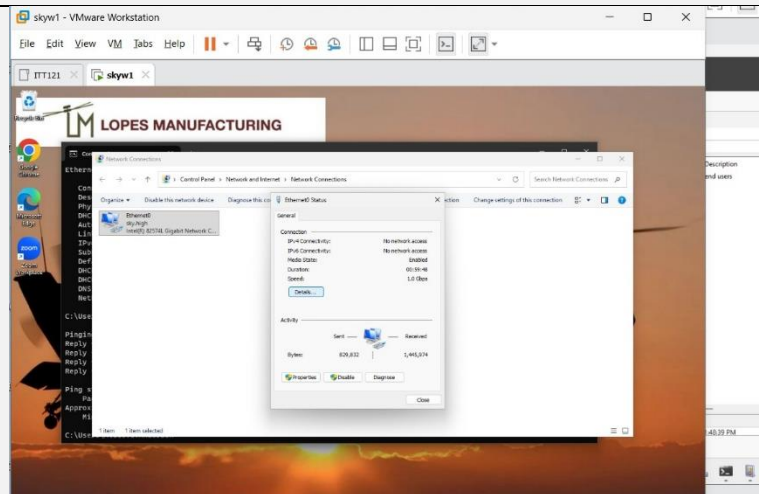


Now switch to the workstation and hit the windows icon to open the search bar. Type out ncpa.cpl and then hit enter.

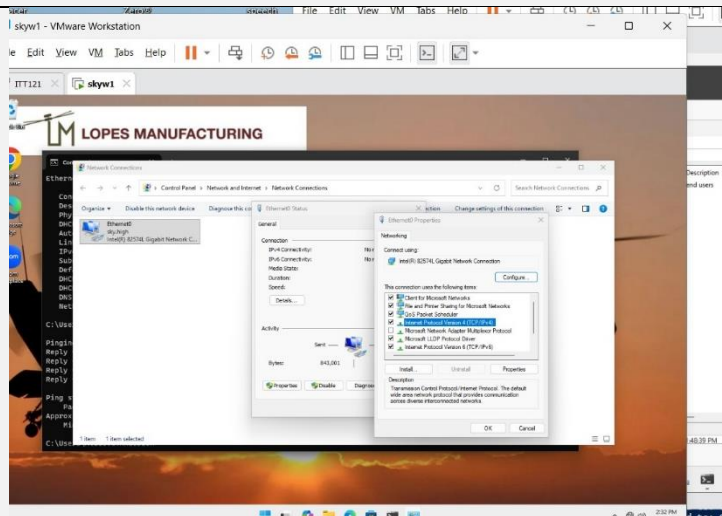


# DHCP

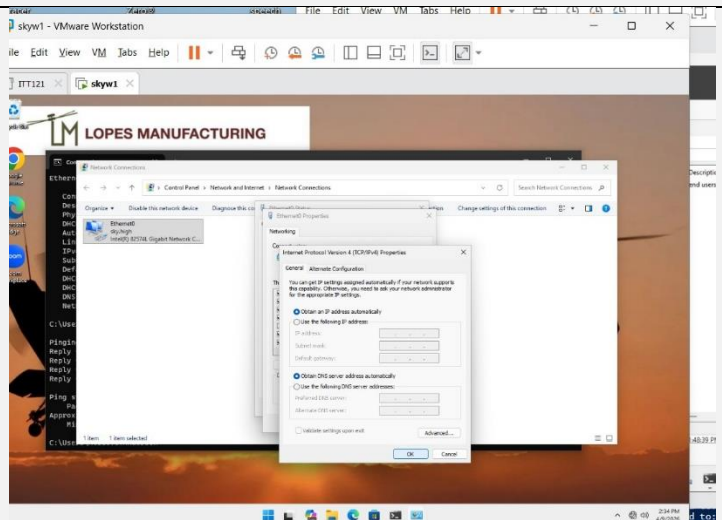
Click on Ethernet when the application opens, and then click on properties.



In the list double click IPv4

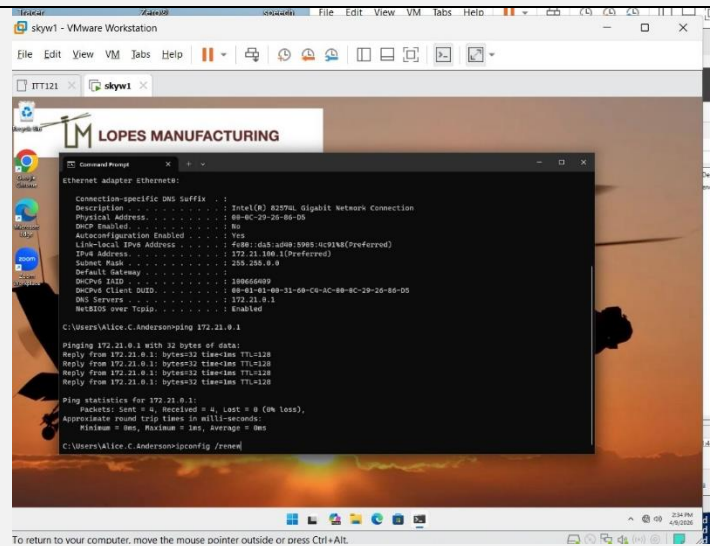


For IP address and DNS server click the box “Obtain IP/DNS automatically”. Hit ok and then close the applications.



# DHCP

Now in the workstation open command prompt by clicking the windows icon, typing cmd in the search bar and then hitting enter on the keyboard. In the terminal type “ipconfig /renew” and then hit enter.



```
ipconfig /renew

Ethernet adapter Ethernet0:

    Connection-specific DNS Suffix . : 
    Description . . . . . : Intel(R) i219LM Gigabit Network Connection
    Physical Address. . . . . : 88-6C-29-20-86-05
    DHCP Enabled. . . . . : No
    Autoconfiguration Enabled . . . . : Yes
    Link-local IPv6 Address . . . . . : fe80::da5:ad08:9985:ac91%3{Preferred}
    IPv4 Address. . . . . : 172.21.186.1(Preferred)
    Subnet Mask . . . . . : 255.255.0.0
    Default Gateway . . . . . : 
    DHCPv6 IPID . . . . . : 1800000009
    DHCPv6 Client DUID. . . . . : 08-01-01-00-31-00-C8-AC-88-2F-26-86-05
    DNS Servers . . . . . : 172.21.0.1
    Netbios over Tcpip . . . . . : Enabled

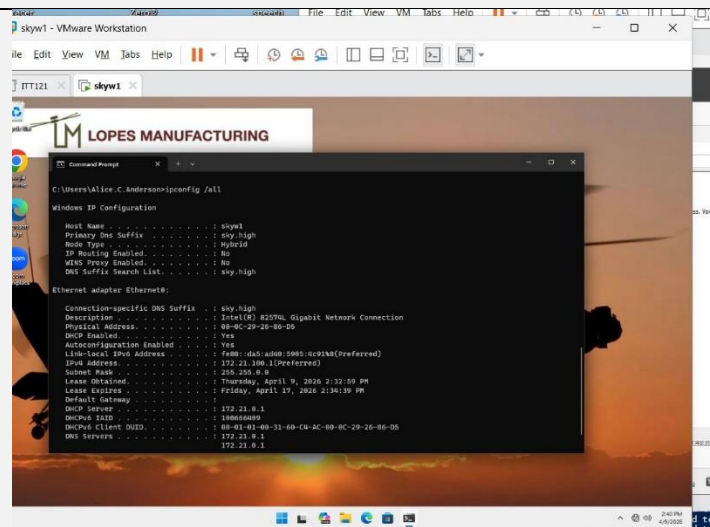
C:\Users\Alice.C.Anderson>ipconfig /renew

Pinging 172.21.0.1 with 32 bytes of data:
Reply from 172.21.0.1: bytes=32 time=1ms TTL=128
Reply from 172.21.0.1: bytes=32 time=1ms TTL=128
Reply from 172.21.0.1: bytes=32 time=1ms TTL=128
Reply from 172.21.0.1: bytes=32 time=1ms TTL=128

Ping statistics for 172.21.0.1:
    Packets: Sent = 4, Received = 4, Lost = 0 (0% loss),
    Approximate round trip times in milli-seconds:
        Minimum = 0ms, Maximum = 1ms, Average = 0ms

C:\Users\Alice.C.Anderson>
```

Now to see our new IP address details, type out “ipconfig /all” in the terminal and hit enter. In the details we can see that DHCP is enabled, and that autoconfiguration is enabled.



```
ipconfig /all

C:\Users\Alice.C.Anderson>ipconfig /all

Windows IP Configuration

    Host Name . . . . . : skyw1
    Primary Dns Suffix . . . . . : sky.high
    Node Type . . . . . : Hybrid
    IP Routing Enabled. . . . . : No
    WINS Proxy Enabled. . . . . : No
    DNS Suffix Search List. . . . . : sky.high

Ethernet adapter Ethernet0:

    Connection-specific DNS Suffix . : sky.high
    Description . . . . . : Intel(R) i219LM Gigabit Network Connection
    Physical Address. . . . . : 88-6C-29-20-86-05
    DHCP Enabled. . . . . : Yes
    Autoconfiguration Enabled . . . . : Yes
    Link-local IPv6 Address . . . . . : fe80::da5:ad08:9985:ac91%3{Preferred}
    IPv4 Address. . . . . : 172.21.186.1(Preferred)
    Subnet Mask . . . . . : 255.255.0.0
    Lease Obtained. . . . . : Thursday, April 9, 2026 2:32:59 PM
    Lease Expires . . . . . : Friday, April 27, 2026 2:34:59 PM
    Default Gateway . . . . . : 
    DHCPv6 IPID . . . . . : 172.21.0.1
    DHCPv6 IPID . . . . . : 1800000009
    DHCPv6 Client DUID. . . . . : 08-01-01-00-31-00-C8-AC-88-2F-26-86-05
    DNS Servers . . . . . : 172.21.0.1
    Netbios over Tcpip . . . . . : Enabled

C:\Users\Alice.C.Anderson>
```

# DHCP

For this class, we don't want to keep DHCP around because if DHCP gets out of our machines we can take out other devices on the network. So in the server, go back to DHCP and delete the scope we just made. Another option is to revert back to a snapshot before we installed DHCP on the server.

